

Memory Layout Diagram

This diagram illustrates the memory organization in the BDI Kernel.

Complete Memory Layout

```
graph TD
    subgraph "Address Space"
        subgraph "Stack Region (High Addresses)"
            A[Stack Top]
            B[Call Frame N]
            C[Call Frame N-1]
            D[...]
            E[Call Frame 0]
            F[Stack Base]
        end

        subgraph "Heap Region (Low Addresses)"
            subgraph "Young Generation"
                G[Eden Space]
                H[Survivor 0]
                I[Survivor 1]
            end

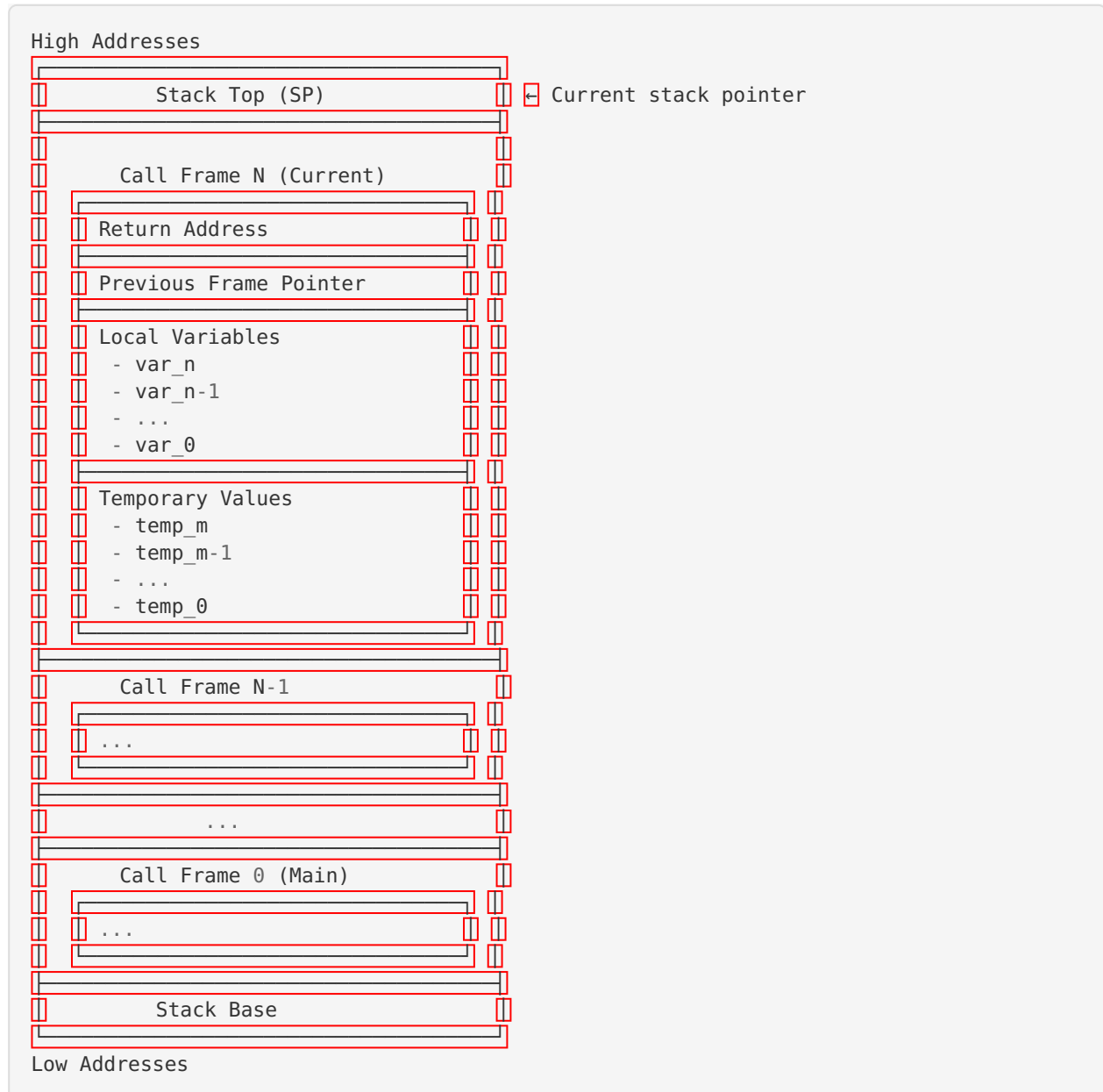
            subgraph "Old Generation"
                J[Tenured Space]
                K[Free List]
            end
        end
    end

    A --> B
    B --> C
    C --> D
    D --> E
    E --> F

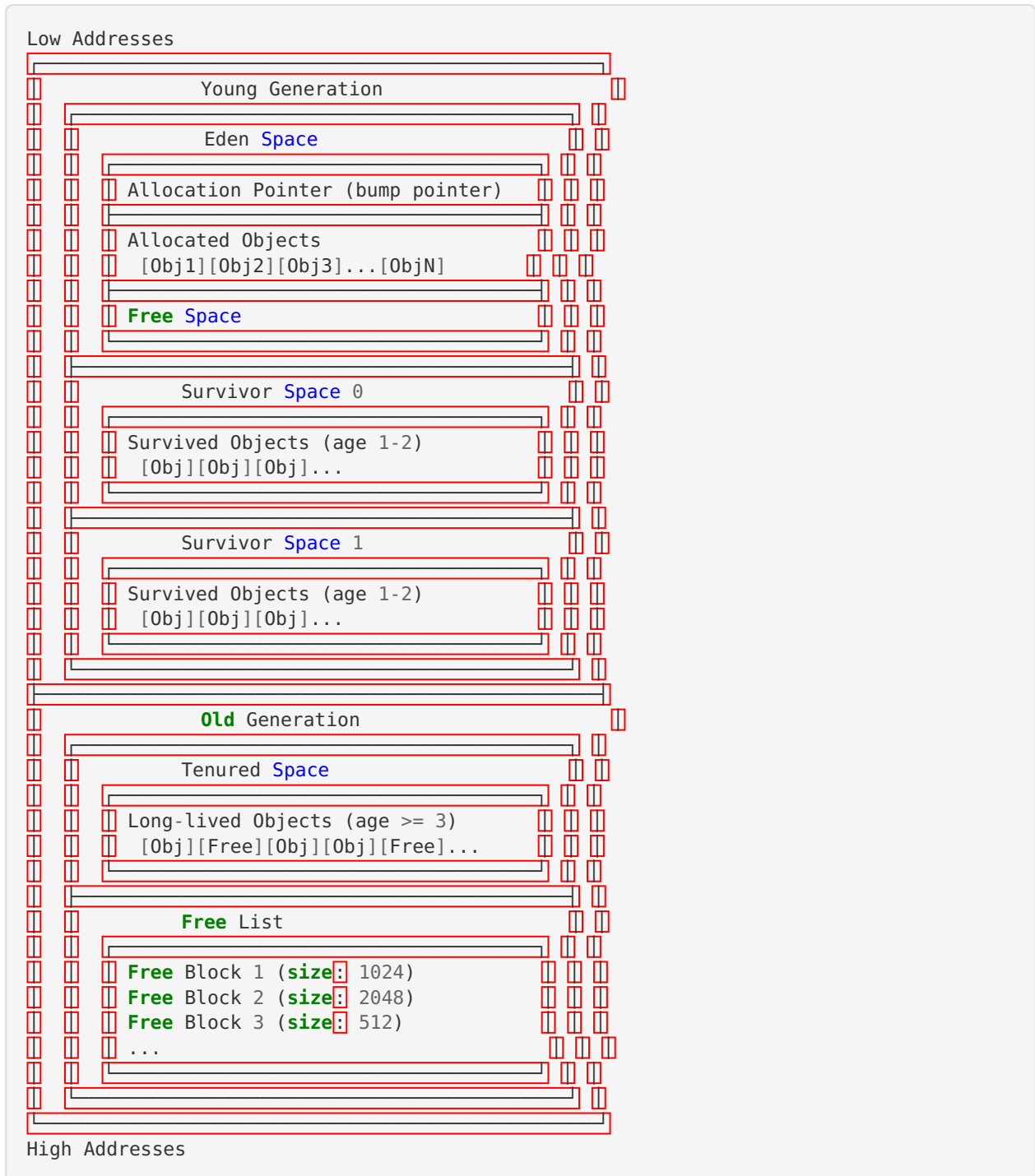
    G --> H
    H --> I
    I --> J
    J --> K

    style A fill:#ffcccc
    style G fill:#ccffcc
    style J fill:#ccccff
```

Detailed Stack Layout

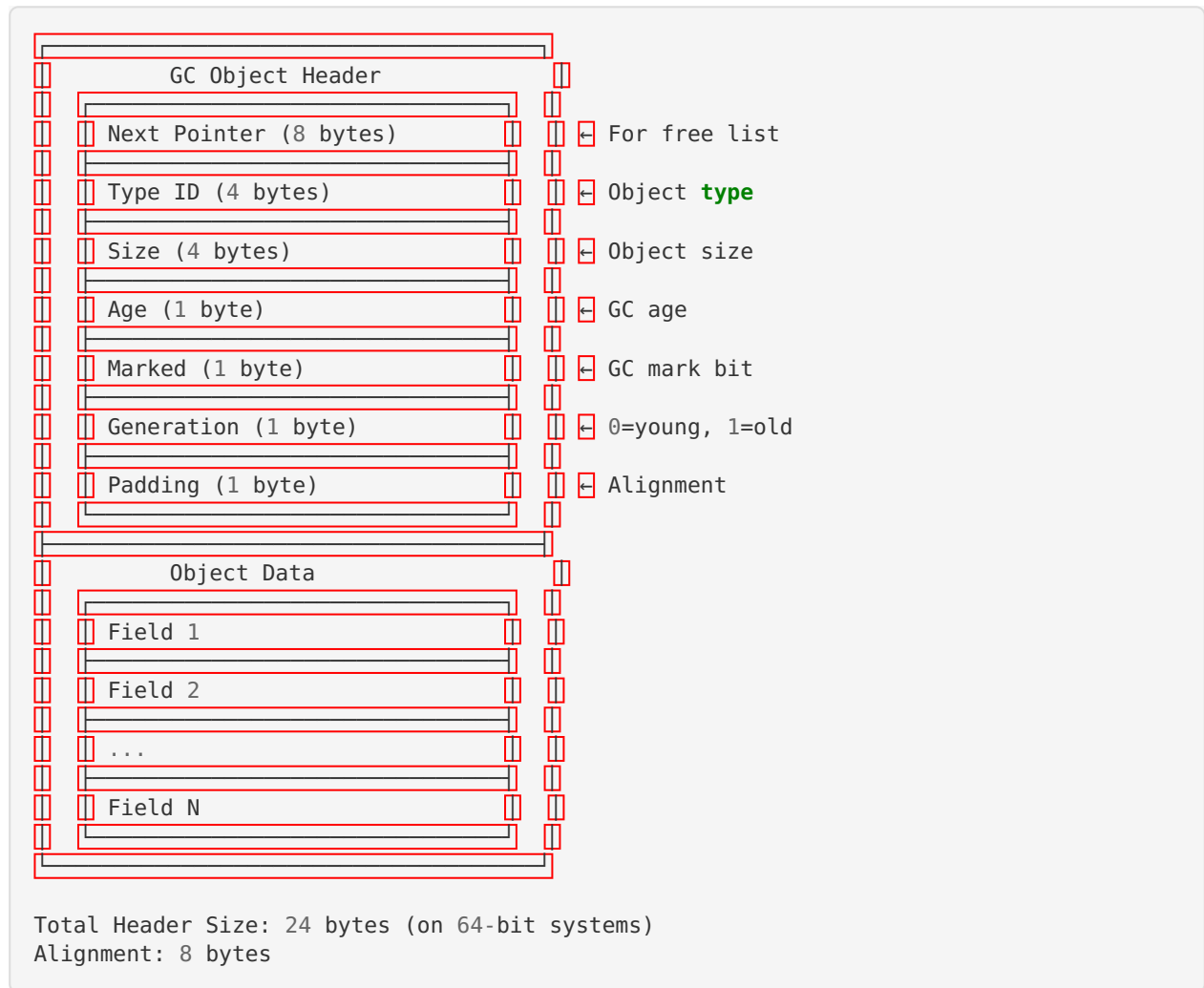


Detailed Heap Layout



Object Memory Layout

Object Header



Memory Allocation Flow

```

flowchart TD
    A[Allocation Request] --> B{Object Size}
    B -->|< 1 KB| C[Young Generation]
    B -->|>= 1 KB| D[Old Generation]
    C --> E{Eden Space Available?}
    E -->|Yes| F[Bump Pointer Allocation]
    E -->|No| G[Minor GC]
    F --> H[Return Object]
    G --> I[Copy Live Objects]
    I --> J[Survivor Space]
    J --> K{Age >= 3?}
    K -->|Yes| L[Promote to Old Gen]
    K -->|No| M[Keep in Young Gen]
    L --> D
    M --> C
    D --> N{Free Block Available?}
    N -->|Yes| O[Free List Allocation]
    N -->|No| P[Major GC]
    O --> H
    P --> Q[Mark and Sweep]
    Q --> R[Compact optional]
    R --> D
  
```

Card Table for Write Barriers

Old Generation Memory:

Card 0	Card 1	Card 2	Card 3	Card 4
512 B	512 B	512 B	512 B	512 B

↓ ↓ ↓ ↓ ↓

Card Table:

C	D	C	C	D
---	---	---	---	---

C = Clean, D = Dirty

Dirty cards contain old-to-young references

Memory Regions and Sizes

Default Configuration

Total Heap: 1 MB

- └ Young Generation: 256 KB (25%)
 - └ Eden: 192 KB (75% of young)
 - └ Survivor 0: 32 KB (12.5% of young)
 - └ Survivor 1: 32 KB (12.5% of young)
- └ Old Generation: 768 KB (75%)
 - └ Tenured Space: 768 KB
 - └ Card Table: ~1.5 KB (0.2% overhead)

Memory Overhead

Component	Size	Percentage
Object Headers	24 B/obj	~10-15%
Card Table	0.2%	0.2%
Free List Metadata	1-5%	1-5%
GC Metadata	1-2%	1-2%
Total Overhead		~12-22%

Memory Access Patterns

Stack Access (Fast)

Access Time: 1-2 CPU cycles
Cache Hit Rate: 95-99%
Pattern: Sequential, predictable

Young Gen Access (Fast)

Access Time: 2-5 CPU cycles
Cache Hit Rate: 80-90%
Pattern: Sequential allocation, random access

Old Gen Access (Moderate)

Access Time: 5-20 CPU cycles
Cache Hit Rate: 60-80%
Pattern: Random access, fragmented